Hyponatremia — Serum Osmolality First (Rule Out Pseudo)

1. Rule out pseudo first

WHAT YOU SEE

Banner: RULE OUT PSEUDO THEN VOLUME THEN URINE Na

WHAT IT ENCODES

The boards algorithm in order: (1) measure SERUM OSMOLALITY to confirm the hyponatremia is truly HYPOtonic (<275 mOsm/kg) — this excludes pseudohyponatremia (isotonic, normal osm) and translocational/hypertonic causes (high osm, e.g., hyperglycemia, mannitol); (2) assess VOLUME status (hypo-/eu-/hypervolemic); (3) check URINE Na (and urine osm). Only hypotonic hyponatremia represents true excess water relative to solute and needs the water-handling workup.

BOARD TRAP

Stem gives Na 128 and the choices jump straight to 'fluid restrict' or 'check volume status' without an osmolality. WRONG first step — without serum osm you can't tell true hypotonic hyponatremia from pseudohyponatremia (lipids/protein) or hyperglycemic translocational hyponatremia, which are managed completely differently. Osmolality is always step one.

2. 2×Na + Glu + BUN

WHAT YOU SEE

Sign: 2 Na PLUS GLU OVER 18 PLUS BUN OVER 2.8 — EXCLUDE PSEUDO

WHAT IT ENCODES

Calculated osmolality = 2×[Na] + glucose/18 + BUN/2.8 (mg/dL units). Compare to the MEASURED osm: a large osmolar gap (measured − calculated > ~10) signals unmeasured solute. Note BUN is an INEFFECTIVE osmole (crosses membranes freely), so it raises measured osm but does NOT cause water shifts or true hyponatremia — tonicity depends on EFFECTIVE osmoles (Na, glucose, mannitol). Use the gap to flag pseudohyponatremia (lipids/paraproteins) or toxic alcohols.

BOARD TRAP

A choice treats a high BUN as if it caused the hyponatremia. WRONG — urea is an ineffective osmole, so uremia raises measured osmolality without shifting water or lowering effective tonicity; it does not produce true hyponatremia or warrant Na treatment. Distinguish osmolality (all solutes) from tonicity (effective solutes only).

3. Hyperglycemia correction

WHAT YOU SEE

Sign: HYPERGLYCEMIA PLUS 1.6 TO 2.4 PER 100, with glucometer

WHAT IT ENCODES

Hyperglycemia causes HYPERtonic (translocational) hyponatremia: glucose is an effective osmole that pulls intracellular water into plasma, diluting Na. Correct the measured Na by adding ~1.6 mEq/L (classic) to ~2.4 mEq/L (more accurate, esp. glucose >400) for every 100 mg/dL of glucose above 100. Measured osm is HIGH, not low. Treating the hyperglycemia (insulin + fluids) brings glucose down and Na rises back on its own.

BOARD TRAP

DKA/HHS patient, glucose 900, Na 128, and a choice gives 3% hypertonic saline for 'hyponatremia.' WRONG — corrected Na ≈ 128 + 2.4×8 ≈ 147 (actually HYPERnatremic). Fix the glucose; giving hypertonic saline risks dangerous overcorrection. Always glucose-correct the Na in any hyperglycemic stem before judging it.

4. Fisherman / pseudo (lipids)

WHAT YOU SEE

Fisherman in yellow raincoat at left of stormy room

WHAT IT ENCODES

Isotonic pseudohyponatremia: severe hypertriglyceridemia or paraproteinemia (e.g., multiple myeloma, Waldenström, IVIG) expands the non-aqueous fraction of plasma. Flame-photometry/indirect ISE methods report a falsely low Na because they assume normal plasma water content. MEASURED serum osmolality is NORMAL (isotonic), so it's a lab artifact, not real water excess. Direct (whole-blood) ISE or blood-gas Na reads correctly.

BOARD TRAP

Milky/lipemic serum, Na 125, normal measured osm; a choice says 'true hyponatremia, restrict fluids / give saline.' WRONG — measured osm is normal, so this is a measurement artifact (pseudohyponatremia); the real Na is normal. No Na-directed treatment; instead treat the lipids/protein and remeasure Na by direct ISE.

5. Hypovolemic branch

WHAT YOU SEE

Sailor labeled HYPOVOLEMIC at left-center

WHAT IT ENCODES

After confirming TRUE hypotonic hyponatremia, assess volume. Hypovolemic = dry: orthostasis, tachycardia, flat neck veins, BUN/Cr >20, HIGH uric acid. Low effective circulating volume turns ADH on via baroreceptors. Treat with isotonic 0.9% NaCl, then watch for autocorrection once volume is restored and ADH switches off.

BOARD TRAP

Calling a dry, orthostatic patient 'SIADH' and restricting fluids. WRONG — hypovolemic hyponatremia needs VOLUME (NS), and restriction worsens it. The euvolemic/SIADH branch is for patients with a normal volume exam, low uric acid, and inappropriately concentrated urine — not the dry patient.

6. Hypovolemic urine Na split

WHAT YOU SEE

Two signs: U Na LESS 20 GI/skin losses, U Na OVER 20 salt-wasting/adrenal

WHAT IT ENCODES

Within hypovolemia, split by urine Na: <20 mEq/L = EXTRARENAL loss (GI vomiting/diarrhea, skin sweat/burns, third-spacing) with appropriately avid kidneys; >20 mEq/L = RENAL loss (thiazides, salt-wasting nephropathy, cerebral salt wasting, adrenal insufficiency, osmotic/RTA). Caveat: vomiting can give a paradoxically high urine Na from bicarbonaturia — use urine Cl (<20) to unmask the volume depletion.

BOARD TRAP

Assuming all hypovolemic hyponatremia is GI/extrarenal. WRONG when urine Na is >20 — that points to a renal/adrenal cause; scan for a thiazide (commonest drug), Addison's (hyperkalemia + hypotension), or CSW (recent CNS event, truly dry). The discriminator is urine Na, refined by urine Cl in the vomiting patient.

7. Euvolemic branch

WHAT YOU SEE

Cloaked figure labeled EUVOLEMIC at center

WHAT IT ENCODES

Euvolemic hyponatremia (normal volume exam): differential is SIADH, hypothyroidism (myxedema), glucocorticoid/adrenal deficiency (secondary), primary polydipsia, and low-solute intake (beer potomania, tea-and-toast). Always exclude hypothyroidism and cortisol deficiency before settling on SIADH, since both are reversible and mimic it. Discriminate the rest with urine osmolality.

BOARD TRAP

Diagnosing SIADH without first checking TSH and a cortisol/ACTH-stim. WRONG — hypothyroidism and adrenal/glucocorticoid deficiency are diagnoses of necessity to exclude; missing them means missing a treatable cause and mislabeling a euvolemic patient as idiopathic SIADH. SIADH is a diagnosis of exclusion.

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8. SIADH high urine osm

WHAT YOU SEE

Sign: Uosm OVER 100, Uosm OVER 100 near euvolemic figure

WHAT IT ENCODES

SIADH: inappropriate ADH → urine is inappropriately CONCENTRATED (Uosm >100, typically >300) and urine Na is usually >30–40, despite low serum osm; clinically EUVOLEMIC with LOW serum uric acid (<4) and low BUN (volume-expanded, increased urate clearance). Criteria: hypotonic hyponatremia, Uosm >100, urine Na >30, euvolemia, normal thyroid/adrenal, no diuretics. Causes: CNS (stroke, hemorrhage), pulmonary (pneumonia, small-cell lung cancer — ectopic ADH), drugs (SSRIs, carbamazepine, cyclophosphamide), pain/nausea/post-op.

BOARD TRAP

Diagnosing SIADH when the urine is DILUTE (low Uosm). WRONG — SIADH cannot make dilute urine; a low Uosm indicates primary polydipsia or low-solute intake instead. Conversely, don't forget LOW uric acid supports SIADH while HIGH uric acid suggests hypovolemia — use it to separate the two euvolemic-looking pictures.

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9. Beer potomania / low solute

WHAT YOU SEE

Sign: BEER POTOMANIA, Uosm LESS 100, LOW SOLUTE LOW INTAKE

WHAT IT ENCODES

Low-solute states (beer potomania, tea-and-toast) and primary polydipsia produce maximally DILUTE urine (Uosm <100) with appropriately SUPPRESSED ADH: the kidney can't excrete water without solute to obligate it (max water clearance = solute load / minimum Uosm). Euvolemic, low urine Na. Treatment is solute/protein repletion — and once solute returns, a brisk water diuresis can overcorrect Na fast (high ODS risk).

BOARD TRAP

Mistaking low-solute hyponatremia for SIADH. WRONG — Uosm is LOW here (kidney diluting maximally), the OPPOSITE of SIADH's concentrated urine; ADH is suppressed, so vaptans are useless. Treat with cautious solute repletion and brace for autocorrection, not fluid restriction + vaptan.

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10. Hypervolemic branch

WHAT YOU SEE

Obese figure labeled HYPERVOLEMIC at right

WHAT IT ENCODES

Hypervolemic hyponatremia: edematous states (heart failure, cirrhosis, nephrotic syndrome, advanced renal failure). Total body Na AND water are both increased, but water is retained in excess because LOW effective arterial volume (poor cardiac output / arterial underfilling) triggers RAAS + ADH. Clinically: edema, ascites, JVD, crackles. First-line: water and salt restriction + loop diuretics; treat the underlying organ failure.

BOARD TRAP

Giving 0.9% NS or generous fluids to an edematous CHF/cirrhotic patient with hyponatremia. WRONG — they are volume-OVERLOADED; saline worsens edema. Despite a high total body volume, effective arterial volume is low (that's why ADH/RAAS are on), so the answer is restriction + loop diuretic, not more fluid.

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11. Hypervolemic urine Na

WHAT YOU SEE

Sign: U Na LESS 20 / SODIUM AVID at right

WHAT IT ENCODES

In hypervolemic hyponatremia, urine Na is LOW (<20) in HF, cirrhosis, and nephrotic syndrome because low effective arterial volume drives intense RAAS-mediated Na avidity (the body hoards Na despite edema). The exception is RENAL FAILURE, where the diseased kidney can't conserve Na, so urine Na is >20.

BOARD TRAP

Seeing edema + low urine Na and calling it 'pre-renal hypovolemia, give fluids.' WRONG — in HF/cirrhosis the low urine Na reflects effective arterial underfilling within an overloaded body; the treatment is restriction + diuresis. Urine Na >20 in an edematous patient instead suggests intrinsic renal failure as the driver.

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12. Correction limit / ODS

WHAT YOU SEE

Brain labeled PONS with red glow, OSMOTIC DEMYELINATION SYNDROME sign

WHAT IT ENCODES

Correct CHRONIC hyponatremia SLOWLY: rise <8 mEq/L per 24h (some use <10), and only ~4–6 mEq/L/24h in high-risk patients. Too-fast correction shrinks brain cells that had osmotically adapted (lost organic osmolytes) → osmotic demyelination syndrome (central pontine myelinolysis), presenting days later with dysarthria, dysphagia, quadriparesis, locked-in syndrome — often irreversible. The acute/seizing exception: give 3% saline to raise Na ~4–6 mEq quickly, then stay within the 24h cap.

BOARD TRAP

Raising Na rapidly in chronic (>48h) hyponatremia 'to normalize it.' WRONG — speed, not the low Na itself, causes ODS; the brain tolerates a stable low Na far better than a fast rise. Cap at <8/24h (4–6 if high-risk), and if you overshoot, actively re-lower with D5W + DDAVP.

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13. DDAVP rescue

WHAT YOU SEE

Small figure labeled DDAVP RESCUE with descending arrow

WHAT IT ENCODES

Overcorrection rescue: if Na rises faster than target, RE-LOWER it. Give DDAVP (synthetic ADH) to clamp the renal free-water diuresis plus D5W (electrolyte-free water) to bring Na back down to a safe trajectory. A proactive 'DDAVP clamp' can also be used up front in high-risk patients to keep correction controlled at 4–6 mEq/24h.

BOARD TRAP

Letting an overcorrection 'ride' because the patient looks fine now. WRONG — ODS is DELAYED (1+ days), so an asymptomatic post-correction patient can still develop it; actively re-lower with DDAVP + D5W when the 24h rise exceeds target rather than waiting.

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14. ODS risk factors

WHAT YOU SEE

Sign: RISK FACTORS FOR ODS — ALCOHOLISM, MALNUTRITION, HYPOKALEMIA, LIVER TRANSPLANT

WHAT IT ENCODES

Highest-risk for ODS: chronic ALCOHOLISM, MALNUTRITION, HYPOKALEMIA, advanced liver disease / liver transplant, and very low Na (<105–120). These brains are osmotically depleted/vulnerable — use the stricter 4–6 mEq/L/24h limit and correct K cautiously (note repleting K also raises Na). Beer potomania and tea-and-toast patients sit squarely in this high-risk group.

BOARD TRAP

Applying the standard <8/24h limit to an alcoholic, malnourished, or hypokalemic patient. WRONG — high-risk patients need an even SLOWER target (~4–6/24h) and proactive DDAVP-clamp planning; the usual ceiling is too permissive for them and risks ODS.
